

Al-Balqa Applied University

3. In PLCs rungs power flows from:

- | | |
|------------------------------------|------------------------------------|
| 3. In PLCs rungs power flows from: | |
| a- The right rail to the left rail | b- The up rail to the left rail |
| c- The down rail to the left rail | d- The left rail to the right rail |

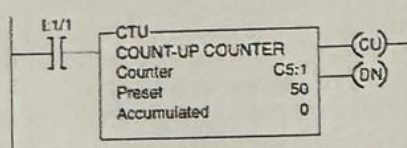
4. Timers may have one or more _____ for their reference timing.

- | | | | |
|---|------------------|-------------------|---------------|
| 4. Timers may have one or more for their reference timing | | | |
| a- Analog outputs | b- Analog inputs | c- Unlatch inputs | d- Time bases |

5. In an up-counter, when the accumulated count exceeds the preset count without a reset, the accumulated count will:

- | | |
|--------------------------------|---------------------------|
| accumulated count will: | |
| a- Hold the accumulated value. | b- Continue incrementing. |
| c -Set itself to zero. | d- Start decrementing. |

6. For the following ladder diagram suppose your accumulated value is 24 and you lose AC line power to the controller. When power is restored to your controller, the counter accumulated value will be:

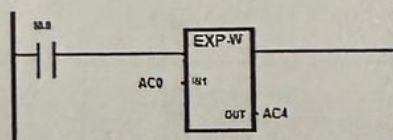


- | | | | |
|------|------|------|-------|
| a-24 | b-26 | c-0. | d-50. |
|------|------|------|-------|

7. The length of time delay period in TOFF timer is calculated by:

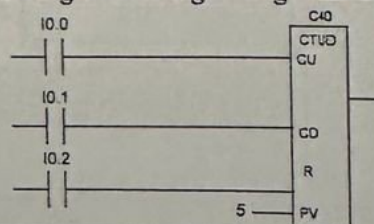
- | | | | |
|--|--|---|--|
| a- The product of the timer time value and the time base | b- The product of the preset value and the timer time. | c- The product of the preset value and the time base. | d- The product of the preset value and the accumulated value |
|--|--|---|--|

8. The instruction list for the following ladder logic diagrams is:



- | | | | |
|------------------------------|-------------------------------|----------------------------------|-------------------------------|
| a- LD I0.0
EXP-W AC4, AC4 | b- LD I0.0
EXP-W AC0, AC0. | c- LD I0.0
EXP-W AC4,
AC0. | d- LD I0.0
EXP-W AC0, AC4. |
|------------------------------|-------------------------------|----------------------------------|-------------------------------|

9. The instruction list for the following ladder logic diagrams is:



- | | | | |
|--|--|---|--|
| a- LD I0.0
CTUD C40, #5
LD I0.1
LD I0.2 | b-LD I0.0
LD I0.1
CTUD C40, 5
LD I0.2 | c- LD I0.0
LD I0.1
LD I0.2
CTUD C40, # 5 | d- LD I0.0
LD I0.1
LD I0.2
CTU C40, # 5 |
|--|--|---|--|

10. A PLC down-counter (CTD) counter counts _____ transitions

- | | | | |
|-----------------|------------------|------------------|-------------------|
| a- true to true | b- false to true | c- false to zero | d- false to false |
|-----------------|------------------|------------------|-------------------|

[illegible]



Al-Balqa Applied University

جامعة البلقاء التطبيقية

Q4 (10 marks)

Write a ladder diagram program using Allen Bradley PLC unit to convert Celsius temperature to Fahrenheit based on the following equation $F^{\circ} = C^{\circ} \times \frac{9}{5} + 32$

The program is designed to convert the recorded Celsius temperature in the data table to Fahrenheit values for display (O:13).